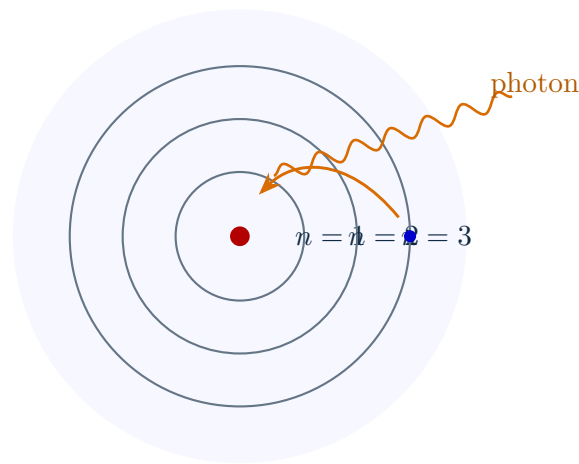


The Bohr Model of the Atom

Quantization and the Origin of Spectral Lines

Lecture notes for computer science students



Main idea

Atoms do not emit arbitrary colors. They emit selected wavelengths because electrons occupy selected energy levels. Bohr's model is a simple bridge between early quantum ideas and the hydrogen spectrum.

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1 What problem are we trying to solve?

Classical physics could describe many continuous systems: planetary motion, waves on a string, electromagnetic waves. But atoms revealed something strange: when hydrogen emits light, it does not produce a continuous rainbow. It produces sharp lines at specific wavelengths.

Question for the lecture

If an atom could have any energy, why would it emit only selected colors?

The answer is the central idea of early quantum theory:

atomic energy is discrete, not continuous.

For this lecture, we focus on the hydrogen atom: one proton and one electron. It is the simplest atom and the first one for which a successful quantum model was proposed.

Learning goals

After this lecture, you should be able to:

- explain why spectral lines suggest discrete atomic energy levels,
- state Bohr's three main postulates,
- derive the allowed radii of the electron orbits,
- derive the energy levels

$$E_n = -\frac{13.6 \text{ eV}}{n^2},$$

- obtain the Rydberg formula from energy-level transitions,
- identify the Balmer series as transitions ending at $n_f = 2$.

2 Spectral lines: the experimental clue

When an excited hydrogen gas emits light, the visible spectrum contains a small number of bright lines. These lines are not random. In 1885, Balmer found that the visible wavelengths of hydrogen obey a simple pattern:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \dots \quad (1)$$

Here:

- λ is the wavelength of emitted light,
- R is the Rydberg constant,
- n is an integer larger than 2.

The more general Rydberg formula is:

$$\frac{1}{\lambda} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \quad n > m, \quad m, n \in \mathbb{N}. \quad (2)$$

Different values of the lower level m give different spectral series.

Series	Final level	Spectral region
Lyman	$m = 1$	ultraviolet
Balmer	$m = 2$	visible
Paschen	$m = 3$	infrared
Brackett	$m = 4$	infrared

Important interpretation

Balmer's formula was first an empirical pattern. Bohr's model explains why this pattern appears.

3 The classical difficulty

A classical electron moving around a proton is an accelerated electric charge. Classical electrodynamics says that an accelerated charge should emit electromagnetic radiation. If the electron emitted radiation continuously, it would lose energy and spiral into the nucleus.

But real atoms are stable. Hydrogen atoms do not collapse.

Classical problem

Classical physics predicts that atoms should be unstable and that emitted radiation should form a continuous spectrum. Experiments show stable atoms and discrete spectral lines.

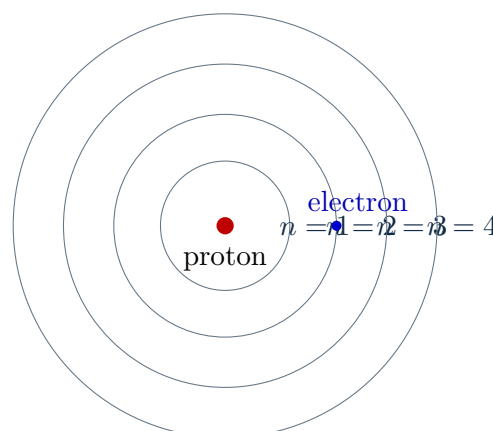
Bohr solved this by keeping part of the classical picture but adding a new quantum rule.

4 Bohr's model: three postulates

The Bohr model is not full modern quantum mechanics. It is a semi-classical model: circular orbits are classical, but the allowed orbits are selected by a quantum condition.

4.1 Postulate 1: only selected orbits are allowed

The electron may move around the nucleus only in certain allowed circular orbits. While the electron remains in one allowed orbit, it does not radiate energy.



4.2 Postulate 2: angular momentum is quantized

The electron's angular momentum is not arbitrary. It must satisfy:

$$L = m_e v r = n \hbar, \quad n = 1, 2, 3, \dots \quad (3)$$

where

$$\hbar = \frac{h}{2\pi}.$$

This is the key quantization condition. The integer n is called the principal quantum number.

4.3 Postulate 3: light is emitted or absorbed during transitions

Light is emitted or absorbed only when the electron changes from one allowed energy level to another.

If the electron moves from an initial level i to a lower final level f , the emitted photon has energy:

$$h\nu = E_i - E_f \quad (4)$$

Since $\nu = c/\lambda$, this can also be written as:

$$\frac{hc}{\lambda} = E_i - E_f \quad (5)$$

Programming analogy

Think of the atom as a system with allowed states. The electron cannot be in arbitrary states. It can occupy only states labeled by $n = 1, 2, 3, \dots$. A photon is emitted when the system changes state.

```
state = n
transition: n_initial -> n_final
photon_energy = E[n_initial] - E[n_final]
```

5 Allowed radii: the minimum derivation

We now derive the allowed orbit radii. We use two ingredients:

1. Coulomb attraction provides the centripetal force.
2. Angular momentum is quantized.

5.1 Step 1: Coulomb force equals centripetal force

For a circular orbit:

$$\frac{k_e e^2}{r^2} = \frac{m_e v^2}{r}. \quad (6)$$

After multiplying both sides by r , we get:

$$m_e v^2 = \frac{k_e e^2}{r}. \quad (7)$$

5.2 Step 2: use Bohr's quantization condition

Bohr's angular momentum condition is:

$$m_e v r = n \hbar. \quad (8)$$

Solving for v :

$$v = \frac{n \hbar}{m_e r}. \quad (9)$$

Substitute this velocity into Eq. (7):

$$m_e \left(\frac{n \hbar}{m_e r} \right)^2 = \frac{k_e e^2}{r}. \quad (10)$$

After simplification:

$$r_n = \frac{n^2 \hbar^2}{m_e k_e e^2} \quad (11)$$

We define the Bohr radius:

$$a_0 = \frac{\hbar^2}{m_e k_e e^2} \approx 5.29 \times 10^{-11} \text{ m} \quad (12)$$

Therefore:

$$r_n = n^2 a_0 \quad (13)$$

Meaning of the result

The atom has discrete allowed sizes. The radius is proportional to n^2 : the second orbit is four times larger than the first, the third is nine times larger, and so on.

6 Energy levels of the hydrogen atom

The total mechanical energy is:

$$E = K + U. \quad (14)$$

The kinetic energy is:

$$K = \frac{1}{2} m_e v^2. \quad (15)$$

The Coulomb potential energy is:

$$U = -\frac{k_e e^2}{r}. \quad (16)$$

From the circular-motion equation:

$$m_e v^2 = \frac{k_e e^2}{r}, \quad (17)$$

so:

$$K = \frac{1}{2} \frac{k_e e^2}{r}. \quad (18)$$

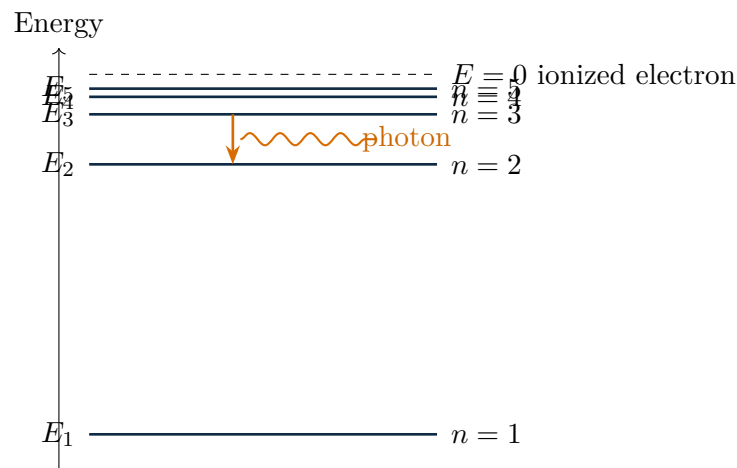
Therefore:

$$E = \frac{1}{2} \frac{k_e e^2}{r} - \frac{k_e e^2}{r} \quad (19)$$

$$= -\frac{1}{2} \frac{k_e e^2}{r}. \quad (20)$$

Using $r_n = n^2 a_0$, we obtain the famous result:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots \quad (21)$$



Why is the energy negative?

The zero of energy is chosen for a free electron infinitely far from the proton. A negative energy means that the electron is bound. To remove the electron from the ground state, we must add 13.6 eV.

7 From energy levels to the Rydberg formula

Now we connect Bohr's energy levels to spectral lines.

Suppose the electron jumps from a higher level n_i to a lower level n_f :

$$n_i > n_f.$$

The photon energy is:

$$\frac{hc}{\lambda} = E_i - E_f. \quad (22)$$

Using:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad (23)$$

we get:

$$E_i - E_f = \left(-\frac{13.6 \text{ eV}}{n_i^2} \right) - \left(-\frac{13.6 \text{ eV}}{n_f^2} \right) \quad (24)$$

$$= 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right). \quad (25)$$

Therefore:

$$\frac{hc}{\lambda} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right). \quad (26)$$

Divide both sides by hc :

$$\boxed{\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)} \quad (27)$$

where approximately:

$$\boxed{R \approx 1.097 \times 10^7 \text{ m}^{-1}}. \quad (28)$$

Main result

Bohr's model explains the Rydberg formula. Spectral lines appear because electrons jump between discrete energy levels.

8 The Balmer series

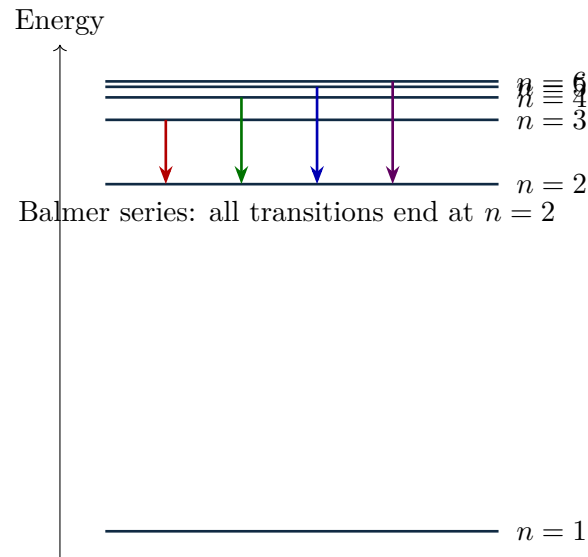
The Balmer series consists of all transitions ending at:

$$n_f = 2. \quad (29)$$

Therefore:

$$\boxed{\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right), \quad n_i = 3, 4, 5, \dots} \quad (30)$$

These transitions produce visible light.



8.1 Example: the red hydrogen line H_α

For the transition:

$$n_i = 3, \quad n_f = 2, \quad (31)$$

we get:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \quad (32)$$

$$= R \left(\frac{1}{4} - \frac{1}{9} \right) \quad (33)$$

$$= R \cdot \frac{5}{36}. \quad (34)$$

Using $R \approx 1.097 \times 10^7 \text{ m}^{-1}$:

$$\lambda \approx 6.56 \times 10^{-7} \text{ m} = 656 \text{ nm}. \quad (35)$$

This is the red line in the visible hydrogen spectrum.

Quick physical reading

Large energy gap means high-energy photon and shorter wavelength. Small energy gap means lower-energy photon and longer wavelength.

9 Emission and absorption

The same energy levels explain two opposite processes.

9.1 Emission

The electron moves from a higher level to a lower level:

$$n_i > n_f.$$

A photon is emitted:

$$h\nu = E_i - E_f.$$

9.2 Absorption

The electron absorbs a photon and moves from a lower level to a higher level:

$$n_i < n_f.$$

The photon must have exactly the right energy:

$$h\nu = E_f - E_i.$$

Why only specific wavelengths?

A photon can be absorbed or emitted only if its energy matches the difference between two allowed atomic energy levels.

10 A compact computational viewpoint

For computer science students, it is helpful to think in terms of states and transitions.

Allowed states:

$$S = \{1, 2, 3, \dots\}$$

Energy function:

$$E(n) = -13.6/n^2 \text{ eV}$$

Transition:

$$n_i \rightarrow n_f$$

Photon energy:

$$\Delta E = E(n_i) - E(n_f)$$

Photon wavelength:

$$\lambda = \frac{hc}{\Delta E}$$

A rough Python-like pseudocode version:

```
def E(n):
    return -13.6 / n**2 # eV
def photon_energy(n_initial, n_final):
    return E(n_initial) - E(n_final)
```

The mathematical model is simple because hydrogen has only one electron. Multi-electron atoms require more advanced quantum mechanics.

11 What Bohr's model explains and what it does not

11.1 What it explains

Bohr's model successfully explains:

- why hydrogen has discrete spectral lines,
- the Balmer formula,
- the Rydberg formula,
- the ionization energy of hydrogen,
- the approximate size of the hydrogen atom.

11.2 What it does not explain

The model has serious limitations:

- it does not work well for atoms with many electrons,
- it treats electrons as particles moving in classical circular orbits,
- it does not include electron spin,
- it does not explain fine structure and many details of spectral lines,
- it is replaced by the wave-mechanical description of quantum mechanics.

Important warning

In modern quantum mechanics, electrons do not literally move on small planetary orbits. The Bohr model is a useful historical and pedagogical model, not the final picture of the atom.

12 Minimal formula sheet

Concept	Formula	Meaning
Bohr quantization	$m_e v r = n \hbar$	only selected orbits are allowed
Bohr radius	$a_0 = \frac{\hbar^2}{m_e k_e e^2}$	size scale of hydrogen
Allowed radii	$r_n = n^2 a_0$	orbit radius grows as n^2
Energy levels	$E_n = -\frac{13.6 \text{ eV}}{n^2}$	discrete bound energies
Photon energy	$h\nu = E_i - E_f$	transition gives light
Frequency-wavelength relation	$\nu = c/\lambda$	wave relation
Rydberg formula	$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$	wavelengths of hydrogen lines
Balmer series	$n_f = 2$	visible hydrogen lines

13 Worked examples

Example 1: Energy of the second level

Find the energy of the hydrogen atom for $n = 2$.

$$E_2 = -\frac{13.6 \text{ eV}}{2^2} \quad (36)$$

$$= -\frac{13.6 \text{ eV}}{4} \quad (37)$$

$$= -3.4 \text{ eV}. \quad (38)$$

Example 2: Energy of the photon for a transition from $n=3$ to $n=2$

$$E_3 = -\frac{13.6 \text{ eV}}{9} \approx -1.51 \text{ eV}, \quad (39)$$

$$E_2 = -\frac{13.6 \text{ eV}}{4} = -3.40 \text{ eV}. \quad (40)$$

The emitted photon energy is:

$$\Delta E = E_3 - E_2 \quad (41)$$

$$= (-1.51 \text{ eV}) - (-3.40 \text{ eV}) \quad (42)$$

$$= 1.89 \text{ eV}. \quad (43)$$

This corresponds to a wavelength of about 656 nm, the red Balmer line.

Example 3: Which transition emits a more energetic photon?

Compare:

$$4 \rightarrow 2 \quad \text{and} \quad 3 \rightarrow 2.$$

Both end at $n = 2$, but $4 \rightarrow 2$ starts higher. Therefore the energy difference is larger, so the emitted photon has higher energy and shorter wavelength.

14 Conceptual check questions

1. Why does the hydrogen atom emit only selected wavelengths?
2. What does the quantum number n label in the Bohr model?
3. Why is the ground-state energy negative?
4. Which transition belongs to the Balmer series: $3 \rightarrow 2$, $2 \rightarrow 1$, or $4 \rightarrow 3$?
5. If a photon has a shorter wavelength, does it have higher or lower energy?
6. Why is Bohr's model useful even though it is not the final model of the atom?

15 Summary

The Bohr model was the first successful model connecting atomic structure with spectral lines. Its key idea is quantization: electrons in hydrogen may occupy only selected energy levels. When an electron moves between these levels, a photon is emitted or absorbed.

The central results are:

$$r_n = n^2 a_0$$

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Final takeaway

The hydrogen spectrum is not random. It is a visible trace of discrete quantum energy levels inside the atom.